Battery Management System (BMS) Final-Year Project

Consolidated English Meeting Minutes

Prepared from the original Chinese meeting record

29 April 2026

Contents

1 Document Control

Project: Battery Management System (BMS) final-year individual project
Student: Yu Zehao / Mark
Supervisor: Dr Wright
Source document: Original Chinese meeting record uploaded as `record Chinese.docx`
Purpose: This document translates the original notes into English and reorganises them into a formal meeting-minutes format. Dates or meeting times that were not explicitly recorded in the source have been retained as week-based entries.

2 Overall Project Direction

The project direction was confirmed as a Battery Management System (BMS) demonstration platform. The project is focused on a practical, low-cost and demonstrable embedded system rather than a commercial-grade BMS. The core technical scope is voltage, current and temperature monitoring, with protection logic and basic user-facing display or data output. More advanced functions, such as SoC/SoH estimation, historical logging, adaptive estimation, balancing strategies and multi-scenario parameter templates, were treated as possible stretch objectives rather than guaranteed baseline deliverables.

3 Consolidated Action Register

Area	Agreed action	Owner	Status / timing
Project definition	Define the BMS scope, target battery type, use case, constraints and deliverable format.	Student	Week 1 onwards
Risk assessment	Prepare a full H&S risk assessment covering thermal runaway, short circuit, over-charge, over-discharge, over-current, over-temperature, mechanical damage, miswiring and operator error.	Student	Required before practical work
Proposal	Submit a concise two-page proposal with aims, objectives, method, milestones, risk and resources.	Student	Deadline: 17 October 2025
Testing strategy	Use a “fail fast, test early” strategy, prioritising early tests of high-risk subsystems.	Student	Throughout project
Temperature measurement	Review required BMS temperature accuracy and improve the sensor placement method.	Student	Early technical risk task
Safety-check logic	Replace the single “Safety Check” block in the flowchart with explicit voltage, temperature and current checks.	Student	Week 6

Area	Agreed action	Owner	Status / timing
PCB planning	Start schematic and PCB work immediately using datasheets and footprints before all parts arrive.	Student	Target: before Christmas break
Procurement	Split normal supplier items and Pi Hut / Adafruit credit-card items into separate purchase emails.	Student / Supervisor	Week 7
Load design	Purchase a 12 V, 21 W automotive bulb and holder for the discharge demonstration.	Student	Week 8
Current sensing	Use a shunt around 0.015 ohm and add input filtering where recommended by the datasheet.	Student	Week 8
INA228 debugging	Do not change the soldered shunt initially; test with a stable known current and distinguish calibration error from noise.	Student / Supervisor	Week 9 / next lab session
Semester 2 planning	Revisit the second-semester plan and update the risk assessment based on reduced and newly emerging risks.	Student	Week 11 onward
Final report	Structure the report around system design, subsystem design, testing, discussion, future work and conclusion.	Student	Semester 2

4 Meeting 1 - Week 1

Date: Friday, 3 October 2025
Participants: Dr Wright; Yu Zehao
Meeting theme: Confirming the project direction, first-week tasks, aims, timeline and deliverables.

4.1 1. Project Scope and Positioning

The project title was confirmed as Battery Management System (BMS). The supervisor advised that the broad concept of "battery management" must be narrowed into a feasible project scope. The project definition should specify the target object, such as a single cell or a small multi-cell lithium-ion battery pack, the intended use case, practical constraints and the final deliverable format.

Because the project must be demonstrated in person, the system should be physically compact and portable. The project is also strongly connected to real applications such as electric vehicles, drones and other battery-powered systems. The final presentation should therefore include a typical application context.

Two possible demonstration routes were discussed:

- modelling or simulation to show the behaviour of algorithms and control strategies;
- a physical demonstration, such as charge, discharge or thermal-control testing using a drone-type battery.

4.2 2. Aims and Objectives

The supervisor advised that the project should separate mandatory functions, advanced functions and staged objectives.

The expected core aims were identified as:

- real-time voltage, current and power monitoring, with display or logging;
- basic state-of-charge (SoC) estimation to provide a usable remaining-energy or runtime indication;
- temperature monitoring and safety-linked control;
- detection of over-temperature, over-current, over-voltage and under-voltage conditions;
- automatic disconnection of charge or discharge paths when over-temperature or other abnormal conditions occur;
- event or alarm logging for safety-related incidents;
- a basic human-machine interface or data interface, such as simple visualisation, serial output or logs.

Possible advanced functions were also discussed, including:

- learning-based or adaptive SoC/SoH estimation using historical charge and discharge data;
- charge/discharge history and reporting, including cycle count and capacity-fade trends;
- strategy optimisation, such as thermal-control logic or a software demonstration of balancing;
- parameter templates for different scenarios, such as drones, small vehicles or energy-storage applications.

The staged objectives should be organised by both time and function. Time-based objectives should define what will be completed each week. Function-based objectives should specify deliverables such as hardware design, software completion and testing.

4.3 3. Safety and Compliance

The H&S risk assessment was identified as a high-priority requirement and had to be completed in the following week. It should include hazards such as thermal runaway, short circuit, over-charge, over-discharge, over-current, over-temperature, mechanical damage, incorrect wiring and human operating error.

The risk assessment should also review relevant literature and common industry practices. It should clearly separate:

- engineering controls, such as fuses, current limiting, thermal control, electrical isolation, protection ICs and emergency disconnection;
- management controls, such as SOPs, PPE and laboratory rules;
- verification methods, especially whether the protective cut-off mechanism works reliably.

The student was also asked to read the project handbook and extract requirements relevant to the project process, documentation, milestones and assessment.

4.4 4. Time and Workload

The estimated project period was around seven weeks or approximately 300 hours, equivalent to roughly two months. Because the available time was limited, the supervisor advised fixing the scope quickly, prioritising basic functions and implementing innovative features only if they did not compromise system stability.

4.5 5. Actions for 6-10 October 2025

The student should:

1. read relevant technical material and identify the required technologies and hardware;
2. draft a project-definition document, including desired functions, mandatory functions and a timeline;
3. prepare a full initial risk assessment, including a hazard list, risk matrix, engineering and management controls and verification methods;
4. extract relevant handbook requirements, milestones and marking criteria;
5. prepare a weekly progress or preliminary presentation;
6. produce an initial proposal covering aims, scope, method, milestones, risk and resource needs.

5 Meeting 2 - Week 2

Date: Wednesday, 8 October 2025
Participants: Dr Wright; Yu Zehao
Meeting theme: Project launch, planning refinement, aims, objectives, motivation and testing strategy.

5.1 1. Core Agenda

The meeting reviewed the initial project plan and refined it based on supervisor feedback. The focus was to improve the clarity and feasibility of the project goals, testing strategy and motivation statement.

5.2 2. Current Project Plan

The project would start from one of the most safety-critical BMS subsystems: temperature management. Once the temperature subsystem was stable, voltage and current management would be integrated progressively. This staged development strategy remained unchanged.

The supervisor also recommended a top-down planning approach. Before detailed design and writing, the student should plan the main content and structure of each part of the work so that the project logic remains clear.

5.3 3. Background

The project background should explain that global electrification has increased the use of batteries across many applications, from electric vehicles to drones. These applications require safer and more efficient battery management.

The BMS should be presented as a core technology for safe battery operation, lifetime extension and performance optimisation. The absence or failure of a BMS can lead to equipment damage and serious safety incidents, creating commercial loss and social risk.

5.4 4. Motivation

The motivation should focus on technical, economic and social value rather than personal interest. A reliable BMS addresses common safety and efficiency problems in battery applications. A successful system can improve product safety and competitiveness, extend battery life and reduce energy loss.

The social motivation is that battery safety becomes increasingly important as battery-powered products become more widespread. The project may help reduce battery-failure incidents and can support the development of safer practice in related industries.

5.5 5. Overall Aim and Objectives

The overall aim was revised to a higher-level statement:

> To develop a BMS prototype that enables safe battery-pack operation through accurate measurement of voltage, current and temperature.

The specific objectives were defined as:

1. **Temperature monitoring and protection:** monitor battery temperature in real time and trigger protective actuation, such as disconnecting the charge or discharge circuit, when the temperature becomes abnormal.
2. **Voltage monitoring and protection:** monitor over-voltage and under-voltage conditions and trigger the corresponding protective action.
3. **Current monitoring and protection:** monitor charge/discharge over-current and short-circuit conditions and trigger the corresponding protective action.

This structure separates the overall aim, which states what the project is trying to achieve, from the objectives, which are specific and verifiable.

5.6 6. Testing Plan

The supervisor recommended moving away from a “perfect plan followed by final testing” model. Instead, the project should use a “fail fast, test early” approach.

The testing strategy should be risk-driven. High-risk technical areas should be tested early at unit or module level, even if the tests are incomplete. The purpose is to expose design defects and integration risks as early as possible.

In electronic product development, rapid iteration is efficient when the cost of iteration in time and money is acceptable. The project should use modular design to control the cost of each iteration.

The first-semester target was to complete independent unit tests of the main subsystems: temperature, voltage and current. The second-semester target was system integration, debugging and overall stress testing.

5.7 7. Risk Assessment

The risk assessment should cover both physical operation risks, such as soldering, charging and assembly, and project-management risks, such as technical difficulty, time pressure and supply-chain delay. The supervisor’s feedback reinforced the importance of identifying high-risk tasks early and testing them quickly.

5.8 8. Actions

The student should continue preparing the Week 3 deliverables. A draft version should be sent to the supervisor before lunch on the following Tuesday where possible. The next meeting was scheduled for 15 October.

6 Meeting 3 - Week 3

Date: Wednesday, 15 October 2025
Participants: Dr Wright; Yu Zehao
Meeting theme: Proposal review, project scope, technical risk and experimental feasibility.

6.1 1. Proposal Review and Scope Definition

The student had prepared an initial BMS proposal. The project direction was to develop a low-cost and efficient BMS prototype with stable operation. The initial development scope focused on three essential BMS measurements: temperature, voltage and current.

The student raised a concern that implementing only these three functions might appear technically too limited for the project. The supervisor confirmed that these functions were a valid and feasible core scope. He introduced the idea of a minimum viable product (MVP): the project should first make the essential functions stable and reliable.

To address the concern about technical depth, the supervisor recommended defining stretch objectives. Advanced functions, such as more complex algorithms or extra sensor integration, could be treated as second-stage goals. This preserves a realistic baseline while leaving space for deeper work if time permits.

The supervisor also emphasised that “cost-effective” must be quantified. The final report should compare the proposed BMS prototype with commercial products in terms of cost and performance. Only quantitative comparison can support the project’s economic value and originality.

6.2 2. Risk Assessment

The student identified coding and software development as a major personal technical risk, because his practical programming experience was not yet fully mature. The supervisor accepted this as a legitimate skill-gap risk and advised including specific time for programming learning and practice in the plan.

The supervisor also identified a deeper technical risk: the accuracy of temperature measurement. If a temperature sensor is simply attached to the surface of the battery pack, the measured value may differ significantly from the internal core temperature of the cell. The difference is caused by thermal resistance and heat-transfer delay, especially during high-rate charge or discharge. This can affect the BMS judgement of the battery safety state.

The supervisor required this temperature measurement and accuracy issue to be included as a key technical risk in the final risk assessment.

6.3 3. Experimental Method and Technical Details

The student initially selected the DS18B20 digital temperature sensor. The supervisor clarified the difference between resolution and accuracy. A sensor may display several decimal places, but this does not mean the reading is close to the true temperature. The DS18B20 can provide high resolution, but its typical absolute accuracy is around +/-0.5 degrees Celsius, which is different from the student's original 0.1 degrees Celsius expectation.

The student should conduct an immediate literature review to determine what temperature-monitoring accuracy is meaningful and accepted in BMS safety and lifetime management. The project specification should then be adjusted based on this evidence.

To improve the physical measurement method, the meeting discussed alternatives to simple surface measurement:

- placing the sensor deeper inside the battery pack, for example in the gap between two cylindrical cells;
- designing a small aluminium or copper fixture that fits closely around several cell surfaces, with the temperature sensor embedded at the centre;
- applying thermal compound between the fixture and the cells to reduce air gaps and thermal resistance.

This structure would improve the response speed and accuracy of the measured temperature by reducing thermal resistance between the cells and the sensor.

6.4 4. Administrative Process and Next Steps

The supervisor clarified that the final proposal should be submitted directly to the official university system by Friday, 17 October. A separate draft review before submission was not required.

The student noted that, under the normal process, component purchasing might not start until the proposal had been formally approved, which would waste one week. The supervisor proposed a practical workaround: the student should email the Health and Safety Risk Assessment directly to him first. The supervisor could review and approve this safety document earlier. Once email approval was received, the student could begin ordering chips, sensors and development boards without waiting for the full system process.

6.5 5. Decisions and Actions

The student should:

1. shorten the proposal to within the required two-page limit;
2. reorganise additional functions as stretch objectives;
3. update the risk assessment to include programming skill gap and temperature measurement accuracy;
4. conduct a literature review to justify the required temperature accuracy;
5. send the H&S risk assessment to the supervisor by email as a high-priority task to enable procurement approval;
6. submit the final proposal by 17 October;
7. begin ordering all required electronic components after receiving the supervisor's safety approval.

7 Meeting 4 - Week 6

Date: Week 6, exact date not recorded
Participants: Student; Supervisor
Meeting theme: Software flowchart review, ACS712 current-sensor accuracy, voltage divider design and PCB manufacturing plan.

7.1 1. Software Flowchart and Pseudocode

The student presented the system software flowchart, which included a block labelled "Safety Check". The supervisor criticised this block as too vague. It was effectively a black box and could not simply return yes or no.

The supervisor advised expanding the block into explicit logic:

1. check voltage and determine whether it is normal;
2. check temperature and determine whether it is normal;
3. check current and determine whether it is normal.

Before writing formal C or C++ code, the student should first write pseudocode or comments. Instead of immediately writing a function such as `void safety_check()`, the student should first define steps such as "read voltage ADC", "convert to float" and "compare with maximum limit". This would clarify the logic and reduce the risk of becoming stuck in syntax.

7.2 2. ACS712 Current Sensor

The student was using the 5 A version of the ACS712 current sensor. The supervisor reviewed the measurement resolution:

- at 0 A, the sensor output is VCC/2, approximately 2.5 V;
- the 5 A version has sensitivity of approximately 185 mV/A;
- the Arduino ADC is 10-bit, with a 0-1023 range and 5 V reference;
- one ADC step is approximately 5 V / 1024 = 4.88 mV;
- a 1 A current change produces approximately 185 mV;
- therefore, the ADC reading changes by approximately 185 mV / 4.88 mV = 37.9 counts per ampere.

The supervisor confirmed that this resolution was sufficient for the project and that current-sensor resolution did not need to be a major concern at this stage.

7.3 3. Voltage Divider

The project needed to measure a 3S LiPo battery pack, with a full-charge voltage of approximately 12.6 V. This must be reduced below the Arduino ADC input limit.

The supervisor warned not to trust resistor colour bands alone. Standard resistors may have 5% or 1% tolerance. For battery measurement, a 5% resistance error could produce a large voltage-reading error, for example reading 12 V as 11.5 V or 12.5 V. This would affect over-charge and over-discharge protection.

The agreed solution was:

1. measure the actual resistance values using a precision multimeter and use the measured values in the code;
2. consider using a potentiometer or trimmer as part of the divider for calibration, adjusting the divider until the software reading matches the multimeter reading;
3. use k-ohm-range resistors, such as 10 k-ohm and 20 k-ohm, to avoid unnecessary battery drain.

7.4 4. PCB Design and Manufacturing

When the student asked when to start PCB design, the supervisor's answer was immediate: start now. Even if components had not arrived, the student could download datasheets and use package dimensions or footprints to create the PCB layout.

The target was to complete the design and submit Gerber files before the Christmas break, around Week 12. This would allow the PCB to be manufactured and delivered during the holiday period, so that soldering and debugging could begin at the start of Semester 2.

7.5 5. Procurement and Prototype Board

The student reported that the PCB laboratory did not have copper foil in stock. The supervisor checked Farnell and concluded that he might need to order it directly.

The student also asked whether it would be acceptable to build a prototype on stripboard or veroboard. The supervisor agreed that breadboard or veroboard could be used for circuit verification, but the final project deliverable should still be a custom PCB.

7.6 6. Actions

The student should:

1. update the flowchart by replacing the single "Safety Check" block with explicit voltage, temperature and current decision logic;
2. begin writing the software framework as comments or pseudocode before detailed syntax;

3. start drawing the schematic in EDA software such as KiCad, Eagle or Altium;
4. create component footprints using datasheets even before physical parts arrive;
5. build and test the voltage-divider prototype, including possible potentiometer calibration.

8 Meeting 5 - Week 7

Date: Week 7, exact date not recorded
Location: Supervisor's office
Participants: Supervisor; Student
Project name used in notes: LiPo Battery Management System Design

8.1 1. Progress Update

The student reported literature research on key battery parameters. The threshold study used references from research papers from other universities. The initial operating temperature range was set to -20 degrees Celsius to +60 degrees Celsius. The student also identified the voltage range for a single cell and noted that a 3-cell pack has a maximum voltage slightly above 12 V.

The hardware progress was:

- current monitoring was planned using a power-monitor module with a built-in sense resistor;
- the LCD module had been soldered and was waiting for testing.

8.2 2. Technical Guidance

The supervisor identified several weaknesses and improvement areas.

First, the data sources needed greater rigour. The supervisor noted that not all LiPo batteries are the same. High-capacity cells for slower charging and high-power cells for fast discharge or fast charging must be distinguished. The student should not apply literature data blindly.

Second, the temperature range must be clearly defined. The student should specify whether the stated range refers to ambient temperature or the surface/internal temperature of the battery. Battery discharge creates self-heating, so these are not identical.

Third, the student had mainly researched discharge or operating conditions. The supervisor required additional research into voltage and temperature thresholds during charging.

8.3 3. Sensor Design and Fail-Safe Behaviour

The supervisor stated that the temperature sensor's measurement range must be significantly higher than the normal operating limit of +60 degrees Celsius. The system should be designed on the assumption that faults can occur. Even during an overheating fault, the sensor must remain capable of reporting data rather than failing together with the rest of the system.

The supervisor also advised adopting a product-design mindset. The design should cover the maximum practical envelope and be suitable for as many battery types as possible, rather than being tied to one specific cell model.

8.4 4. Power and Thermal Management

The supervisor noted that a charging current around 5 A, equivalent to roughly 2C to 3C in the discussed context, is significant. The student should read the MOSFET datasheet, find its R_DS(on), calculate heat generation using P = I^2R and decide whether a heat sink is required.

8.5 5. Procurement Delay

The order submitted two weeks earlier, including the power monitor, had not arrived. The supervisor checked the issue and found that he had not submitted the order to Operations. There was also confusion because he believed the student did not need the copper-foil item. The order included Adafruit / Pi Hut products, which required a credit-card purchase route rather than a normal B2B invoice process.

The agreed solution was to resubmit split orders.

8.6 6. Laboratory Equipment Issue

The student reported that the laboratory soldering station was difficult to use and that solder was not melting properly. The supervisor acknowledged that equipment condition and operating instruction might both be issues. If the student encountered further soldering difficulties, the supervisor offered to take him to a research laboratory and supervise the use of better equipment.

8.7 7. Project Strategy

The supervisor advised against assuming that the project workload can be spread evenly across all weeks. The early weeks of the semester should be used intensively while the course workload is lower.

A lead-time strategy was agreed: before the Semester 1 exam period, around Week 12, the student should complete and order the PCB, enclosure or test fixture. January, when the student is busy with exam preparation, can then be used as the manufacturing and delivery period. This would allow Semester 2 to begin directly with debugging, rather than waiting for custom parts.

8.8 8. Actions

Immediate student actions:

1. go to the PCB laboratory and ask Derek, Gary or Lia whether 0.12 mm copper foil is available;
2. if available, collect it directly, even if the material is held under the supervisor's name;
3. send two procurement emails to the supervisor:
    - Email A: normal order containing most parts, excluding Pi Hut items and excluding copper foil if it has been found;
    - Email B: credit-card order containing only the Adafruit board from Pi Hut.

Tasks for the next week:

1. complete the research on charging-state voltage and temperature thresholds;
2. calculate MOSFET heating at 5 A and determine the required heat-management method;
3. test the assembled LCD screen;
4. begin planning which parts need to be ordered before the end of Semester 1.

Supervisor commitments:

1. send the student the credit-card purchase form template;
2. after receiving the student's emails, forward the order to Operations on the same day.

The next meeting was planned for the same time the following Friday.

9 Meeting 6 - Week 8

Date: Week 8, Friday morning inferred from notes
Location: Supervisor's office, Manchester
Participants: Student; Supervisor
Meeting theme: Hardware progress, load selection, current sensing and final demonstration strategy.

9.1 1. Opening and General Discussion

The student noted that the weather outside was cold. The supervisor commented that he liked the dry, crisp winter weather and mentioned that he is a skier and climber. This was a short informal opening and did not affect technical decisions.

9.2 2. Administration and Applications

The student confirmed that his UCL application had been submitted. The supervisor stated that he had completed the reference for that application. The student planned to submit the University of Edinburgh application the following week, and the supervisor confirmed that this was the only remaining pending application.

9.3 3. LCD Display Debugging

The student reported that the LCD screen was now working. The previous screen did not display anything and appeared to be faulty, so the student bought a replacement for around 4-5 pounds. The new screen had been connected to the temperature-measurement system and was functioning normally.

9.4 4. Temperature Measurement and Data Analysis

The student had measured the NTC thermistor temperature-resistance curve. The curve fit was good, except for one slightly deviating point.

The supervisor advised that if the measured and theoretical curves are very close, plotting them on the same graph may not clearly show the difference. He recommended plotting a residual graph, showing measured value minus theoretical value or a ratio between the two. This would display the error range more clearly and appear more professional in the report.

9.5 5. Load Selection: Bulb vs Resistor

The next stage required a load for a discharge-current demonstration. The discussed battery-pack capacity was approximately 2.2 Ah, or around 2 Ah, and the target discharge current was approximately 2 A. This was considered a safe discharge rate and far below the 4C maximum of about 8 A.

The student initially considered using a high-power resistor. The supervisor recommended using an automotive bulb instead, because it provides a clear visual demonstration while a resistor only produces heat.

The agreed specification was a 12 V, 21 W automotive brake or indicator bulb. Its current is approximately 21 W / 12 V = 1.75 A, which is close to the 2 A target.

The student found that Amazon delivery was too slow. The supervisor recommended suppliers such as CPC or Farnell, and suggested buying a bulb holder with a bayonet fitting to make wiring easier.

9.6 6. Current-Sensing Chip and Circuit Design

The meeting discussed the INA219 current and power monitor. The notes also record that a previous discussion referred to the INA228, so this point may reflect either a comparison between devices or mixed terminology. The technical discussion focused on reading the datasheet and designing the current-sense path.

The supervisor noted that the monitor required a shunt resistor. The full-scale input voltage was very small, around 40-50 mV. To achieve a suitable reading at 3 A without exceeding the range, R = V/I gives approximately 0.015 ohm, or 15 milliohms.

The power dissipation would be I^2R = 3^2 x 0.015 = 0.135 W. Therefore, a 1 W or 0.5 W resistor would be sufficient.

The supervisor also identified a recommended input filter in the datasheet. This involved two 10 ohm series resistors between the shunt and the chip pins, with a capacitor connected for filtering. The purpose was to reduce noise caused by current fluctuations and prevent unstable readings. The supervisor strongly recommended including this filter.

9.7 7. Component Package Choice

The student asked whether to buy surface-mount or through-hole components. The supervisor advised that, for hand soldering, 1206 or 0805 SMD components are usually manageable.

9.8 8. Final Demonstration Strategy

For the final demonstration, the supervisor recommended bringing two bench digital multimeters. One should measure voltage and the other current, allowing a live comparison with the student's system readings. This would help demonstrate measurement accuracy.

9.9 9. Actions

The student should:

1. purchase a 12 V, 21 W automotive bulb and holder;
2. add the input-filter network, consisting of 10 ohm resistors and a capacitor, to the schematic;
3. use a shunt resistor of approximately 15 milliohms;
4. prepare for the next meeting on Friday 28th at 12:00, because the supervisor would be in Scotland on Wednesday.

10 Meeting 7 - Week 9

Date: Thursday or Friday afternoon, exact date not recorded
Location: Supervisor's office, University of Manchester
Participants: Supervisor; Student
Meeting theme: PRM preparation, INA228 prototype review and low-current measurement debugging.

10.1 1. Opening and Workload Update

The supervisor made a brief comment about the automatic office lights being slightly over-intelligent because they remained on for some time even after no one was present.

The supervisor asked about the student's week. The student replied that it was acceptable. The supervisor commented that being able to say "acceptable" usually meant the week was fine, while a truly bad week would feel like moving backwards.

The student explained that he had completed another laboratory task and laboratory report during the week. Therefore, project progress was slow during the first two days, but he had started catching up again from Wednesday.

10.2 2. Administration and PRM Preparation

The supervisor emphasised that the Progress Review Meeting (PRM) should not only show what had been completed in the current week. The better strategy was to review all progress since the previous PRM. This would make the presentation more complete, coherent and likely to score higher.

10.3 3. Hardware and Measurement Review

The student showed the current INA228 prototype measurement system. The student had manually verified that P = V x I matched the chip output. This confirmed that the I2C communication path and the chip's internal measurement logic were operating correctly.

10.4 4. Low-Current Measurement Instability

The student reported that small-current readings, such as around 10 mA, fluctuated significantly. He suspected that the 0.015 ohm shunt resistor might be too small and was concerned that it had been soldered onto the board, making replacement difficult.

The supervisor checked the INA228 parameters and pointed out that the chip has a 20-bit ADC. If the full-scale current were 2 A, the maximum shunt voltage would be 2 A x 0.015 ohm = 30 mV. A 20-bit ADC divides this voltage into about one million levels, giving microvolt or nanovolt-level resolution.

The conclusion was that the hardware sensitivity was more than sufficient. The resistor did not need to be changed. The fluctuation was more likely caused by software range configuration or system noise rather than a hardware defect.

10.5 5. Next Technical Milestone: SoC

The student planned to use the INA228 charge-accumulation function for coulomb-counting-based SoC estimation. The supervisor approved this direction and considered it a reasonable next step.

10.6 6. Debugging Plan

The agreed debugging approach was:

1. do not modify the soldered shunt resistor;
2. use a precision laboratory power supply or source meter to inject a stable known current, for example 100 mA;
3. compare the stable input current with the software reading;
4. if the reading is stable but offset, treat the issue as calibration;
5. if the reading still fluctuates strongly, investigate software configuration or noise floor.

10.7 7. Next Meeting

The next meeting was scheduled for Monday at 10:00 AM. The student would meet the supervisor at the supervisor's office and then go to the Dry Lab for debugging.

The student should bring:

- laptop;
- NUCLEO-G491RE board;
- INA228 prototype.

The supervisor encouraged the student to maintain the current working rhythm and stated that they would identify and solve the measurement-noise issue together on Monday.

11 Meeting 8 - Week 10 Progress Review Meeting

Date: Week 10, exact date not recorded
Meeting type: Progress Review Meeting
Meeting theme: Review of progress from Week 5 to Week 10.

11.1 1. Progress Reported

The student reported progress from Week 5 to Week 10. The main areas were:

- NTC thermistor calibration;
- implementation of the temperature system;
- implementation of the voltage and current measurement system;
- initial MOSFET and LED control functions;
- assembly of a draft prototype;
- preliminary implementation of basic functions.

The student also showed a demonstration video of the draft prototype.

11.2 2. Supervisor Feedback and Marks

The supervisor gave positive feedback on the demonstration video. In the marking discussion, the supervisor indicated that the student's engagement and self-motivation were good and could receive a mark of 8. For technical progress and deliverables, the mark was indicated as 7.

11.3 3. INA228 Issue and Follow-Up

The supervisor also referred to the current problem with the INA228 power monitor. A hardware-check session was arranged for the following Tuesday at 9:30 AM in the laboratory. If no hardware issue was found, the student could then focus mainly on software.

11.4 4. Immediate Tasks

The main tasks for the following two weeks were:

1. complete the INA228 deliverable;
2. complete the draft introduction;
3. complete the draft methodology.

12 Meeting 9 - Week 11

Week: Week 11
Meeting type: Project progress meeting
Format: Short discussion
Meeting theme: End-of-semester planning, system integration and Semester 2 preparation.

12.1 1. Progress Overview

All main subsystems, including measurement, display and control, had completed independent testing. The system was ready for integration. Overall Semester 1 progress was strong, and some tasks had moved faster than originally planned.

12.2 2. Plan for the Remaining 1.5 Weeks

The next priority was full system integration. The completed subsystems should be combined and tested as a complete system. The student should verify stability and functional consistency during full operation.

The meeting also discussed the UKF-based SoC algorithm. The student should reconsider whether the Unscented Kalman Filter (UKF) is necessary and how it should be implemented. If abbreviations such as UKF are used in the demonstration or report, their full names and short definitions must be provided.

12.3 3. Christmas Holiday Plan

Communication and feedback would be possible between 5 January and 8 January. The supervisor recommended completing the PCB design before the Christmas break and submitting the PCB design files for review during 5-8 January.

For the battery solution, because more options were available in China, the student could consider selecting and purchasing the batteries there.

12.4 4. Semester 2 Planning

The student should review and adjust the overall Semester 2 plan. Since several Semester 1 tasks had moved faster than planned, the Semester 2 objectives could be made more targeted and appropriately more ambitious.

The risk assessment should also be updated. Some previously high risks had now been reduced, while new or increased risks might need to be added. The corresponding mitigation measures should be adjusted.

12.5 5. Actions

The student should:

1. complete system-level integration and basic verification;
2. reassess the role and necessity of the UKF SoC algorithm and define terminology clearly in the documentation;
3. complete PCB design before the Christmas break;
4. update the Semester 2 plan and risk assessment.

13 Meeting 10 - Semester 2 Week 1

Week: Semester 2, Week 1
Meeting theme: Holiday progress, PCB fabrication, load components and preparation for live assembly.

13.1 1. Holiday Progress

The student reported project progress made during the holiday period:

- the PCB design had been completed and manufactured;
- several bulbs had been purchased for use as loads;
- two triangular battery packs had been purchased for later experiments and demonstration.

13.2 2. Battery Temperature Measurement

The meeting discussed the battery temperature-measurement arrangement, focusing on how the thermistor could be placed and inserted into the battery structure. An initial feasible implementation method was formed.

13.3 3. Next Arrangement

A live session was scheduled for the following Monday to solder and assemble the PCB.

13.4 4. Actions

The next main action was to carry out PCB soldering and physical assembly on the following Monday.

14 Meeting 11 - Semester 2 Week 3 and Week 5 Notes

Week: Semester 2, Week 3 and Week 5 summary notes
Meeting theme: Progress discussion and future work arrangement; demonstration-related issues.

14.1 1. Week 3 Discussion

The Week 3 discussion covered the current project progress and the work plan for the following weeks. The emphasis was on checking whether the project was moving in the expected direction and identifying the next blocks of work.

14.2 2. Week 5 Discussion

The Week 5 discussion focused on issues related to the project demonstration. The specific details were not fully recorded in the source notes, but the meeting was concerned with how to present the system effectively.

15 Meeting 12 - Semester 2 Week 6

Week: Semester 2, Week 6
Meeting theme: Adding technical depth and structuring the final report.

15.1 1. Technical Development Direction

The meeting mainly discussed two areas. First, the supervisor considered that the overall project was progressing well and that the basic functions had already taken shape. Therefore, the next stage could add some new technical content to improve the project's technical depth and demonstration quality.

The supervisor suggested adding more visual or analytical features, such as:

- a battery-level display box in the interface;
- a battery usage curve;
- a trend display showing how battery parameters change over time.

The main intention was not to redesign the whole project, but to add features on top of the existing system that demonstrate engineering thinking and technical complexity.

15.2 2. Final Report Structure

The second part of the meeting reviewed the overall writing strategy for the final report. The supervisor emphasised that the abstract should be written after the rest of the report has been completed.

The first half of the report could begin with the introduction and project scope. The review section should not simply list which papers were read or what each paper said. Instead, it should show the student's analysis: what was learned from the literature or existing solutions, what problems were identified, and how those findings support the project design.

The main body of the report should be structured around the system and its subsystems. One chapter could cover overall system design, followed by specific sections or chapters on the temperature subsystem, voltage and current measurement subsystem and other parts. The report should then discuss overall testing.

The later part of the report should include discussion, future work and conclusion. References and appendices should appear at the end. Important supporting material that is useful but not central to the main argument should be placed in the appendix.

16 Closing Summary

Across the meetings, the project evolved from a broad BMS idea into a low-cost embedded BMS demonstration platform. The agreed core deliverables were voltage, current and temperature monitoring, safety-related control, display or logging, and system-level integration. The supervisor repeatedly emphasised practical risk reduction, early testing, clear distinction between core objectives and stretch objectives, quantification of cost-effectiveness and professional reporting.

The main technical risks identified were temperature-measurement accuracy, current-measurement noise, correct sensor calibration, thermal design for MOSFET operation, procurement delays and the student's programming skill gap. The main project-management direction was to complete custom hardware planning early enough to use manufacturing lead time during the holiday period, then use Semester 2 for integration, debugging, final testing and report writing.